Homework 6

Submission Instructions

Print this homework and complete it by hand. Scan your completed work as a single PDF and upload it to Brightspace. Loose-leaf paper is also acceptable. Please show all work clearly.

Part 1: Power Series and Functions

1. Find the interval of convergence of the given power series.

(a) $\displaystyle\sum_{k=0}^{\infty} \frac{x^k}{k+2}$

(b) $\displaystyle\sum_{k=2}^{\infty} \frac{x^k}{\ln(k)}$

(c) $\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^k}{k!}(x-3)^k$

(d) $\displaystyle\sum_{k=1}^{\infty} \frac{k(x-2)^k}{e^k}$

(e) $\displaystyle\sum_{k=2}^{\infty} \frac{(k(7x+1)^k}{2^k}$

2. Given that

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad , \text{for } x \in (-1,1).$$

Find the power series for each function centered at $a = 0$ and state its radius of convergence.

(a) $f(x) = \dfrac{1}{1-2x}$

(b) $f(x) = \dfrac{1}{1+4x^2}$

(c) $f(x) = \dfrac{x}{1+x^2}$

(d) $f(x) = \dfrac{x-1}{x+1}$

Part 2: Differentiation and Integration of Power Series

1. Find a power series representation for the given function by differentiating a known power series. State the radius of convergence and interval of convergence.

   (a) $f(x) = \dfrac{2}{(1+x)^2}$ $\quad \left(Hint: f(x) = -2\dfrac{d}{dx}\left(\dfrac{1}{1+x}\right)\right)$

   (b) $f(x) = \dfrac{x}{(1+x^2)^2}$

   (c) $f(x) = \dfrac{1-x^2}{(1+x^2)^2}$ $\quad \left(Hint: f(x) = \dfrac{d}{dx}\left(\dfrac{x}{1+x^2}\right)\right)$

2. Find a power series representation for the given function by integrating a known power series. State the radius of convergence and interval of convergence.

   (a) $f(x) = \ln(1 + x)$

   (b) $f(x) = \ln(1 - x)$

   (c) $f(x) = x \ln(1 + x)$

   (d) $f(x) = \arctan(2x)$

Part 3: Taylor and Maclaurin Series

1. Find the Taylor or Maclaurin series for the given function expanded about the given point and determine the numbers x for which the series converges.

   (a) $f(x) = e^{2x}, \qquad a = 0$

   (b) $f(x) = 1 + x^2, \qquad a = 2$

   (c) $f(x) = \ln(3 + x), \qquad a = 0$

   (d) $f(x) = \sqrt{x}, \qquad a = 4$

2. Find the n-th Taylor Polynomial for the function f expanded about $x = a$.

(a) $f(x) = e^{-x}, \qquad a = 0, \qquad n = 4$

(b) $f(x) = \cos(x), \qquad a = \pi/4, \qquad n = 6$

(c) $f(x) = \dfrac{1}{1+x^2}, \qquad a = 0, \qquad n = 2$

(d) $f(x) = \sqrt{1-x}, \qquad a = 0, \qquad n = 3$

Part 4: Parametric Equations

1. Sketch the curves below by eliminating the parameter t and find an equation in $xy-$coordinates whose graph contains the given curve.

   (a) $x(t) = t, \quad y(t) = t + 6$

   (b) $x(t) = t^2 + 2t, \quad y(t) = t + 1$

   (c) $x(t) = e^t, \quad y(t) = e^{2t}$

   (d) $x(t) = \sin(t), \quad y(t) = \cos^2(t)$

2. (a) Find the parametric equations for the graph $y + \sqrt{x} = e^{2x}$.

(b) Find the parametric equations for the line containing the point $P_0 = (2, 4)$ and with slope 3.

(c) Find parametric equations for the ellipse $\frac{x^2}{7} + \frac{y^2}{13} = 1$.

3. A particle moves in the plane so that at time t its position is given by $x(t) = t + 4, y(t) = 8 - t^2$. A second particle moves in the plane so that at time t its position is given by $x(t) = t + 4, y(t) = t + 6$.

(a) Find equations in $xy-$coordinates for each of the curves.

(b) Do the paths of the particles cross? If so, where?

(c) Do the particles collide? If so, where and at what time?